Aperture-coupled microstrip antenna with FSS superstrate layer at 110 GHz

Lower substrate: For the first antenna topology (Antenna1), Fig. 1 shows a h-slot microstrip patch fed by a microstrip line that is designed and simulated. The patch dimension, slot dimension and the substrate thickness and permittivity is designed for optimum coupling.

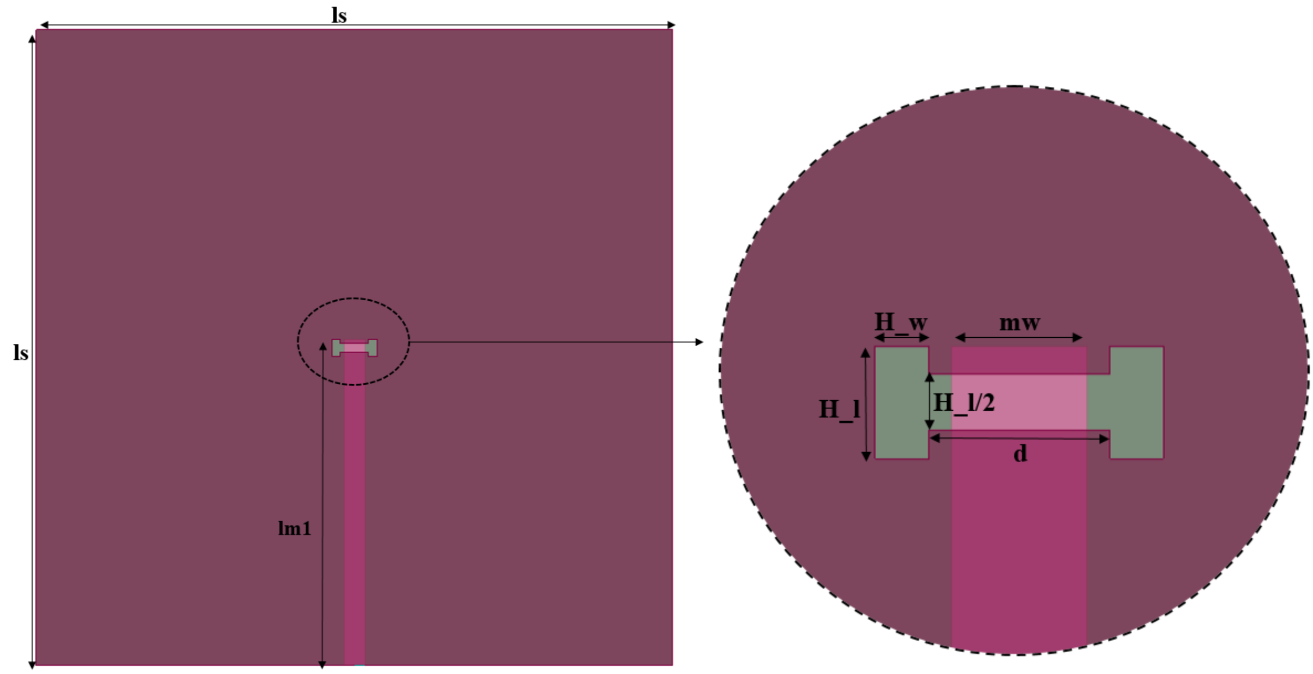


Fig.1. H-slot on ground plane

The substrate, having a relative permittivity of 3 has dimensions of ls=7.36 mm in length, 7.36 mm in width, and 0.064 mm in thickness.

1. **Ground Plane with H-Shaped Slot**: The top surface of the substrate has a conductive ground plane with an H-shaped slot cut into it.
2. **Microstrip Line**: A microstrip line is visible on the bottom surface of the substrate. This line is used to feed energy to the ground plane and radiate the H-shaped slot.
3. **Dimensions and Alignment**: The H-slot is centrally located and symmetrically aligned with respect to the ground plane. The H-shaped slot in the ground plane has specific dimensions: the central horizontal section of the slot is 0.1023 mm wide and 0.3273 mm long, connecting two vertical rectangular sections on either side. Each vertical section is 0.2045 mm long and 0.0972 mm wide. The structure is symmetrical about its centre and aligns with the 0.2455 mm wide microstrip line from the substrate's bottom surface, ensuring efficient coupling or resonance characteristics. The dimensions in the figure are specified in table 1.

Table 1: Dimensions of Fig.1

Parameters	**Description**	**Dimensions (mm)**
ls	Length of substrate	7.36
lm1	Length of microstrip line	3.78
H_l	Length of side arm of H-slot	0.2045
H_w	Width of side arm of H-slot	0.0972
mw	Width of microstrip line	0.2455
d	Horizontal distance between two arms	0.3273

The microstrip line is connected to the ground plane via the feed line, made of a perfect conductor, which has a height same as that of the substrate and width of 0.2455 mm.

Middle substrate: For the second antenna topology (Fig.2), shows a single element rectangular radiating patch that is designed and simulated on the bottom face of the substrate with relative permittivity of 3 and thickness of 0.0639mm (Rogers RO3003).

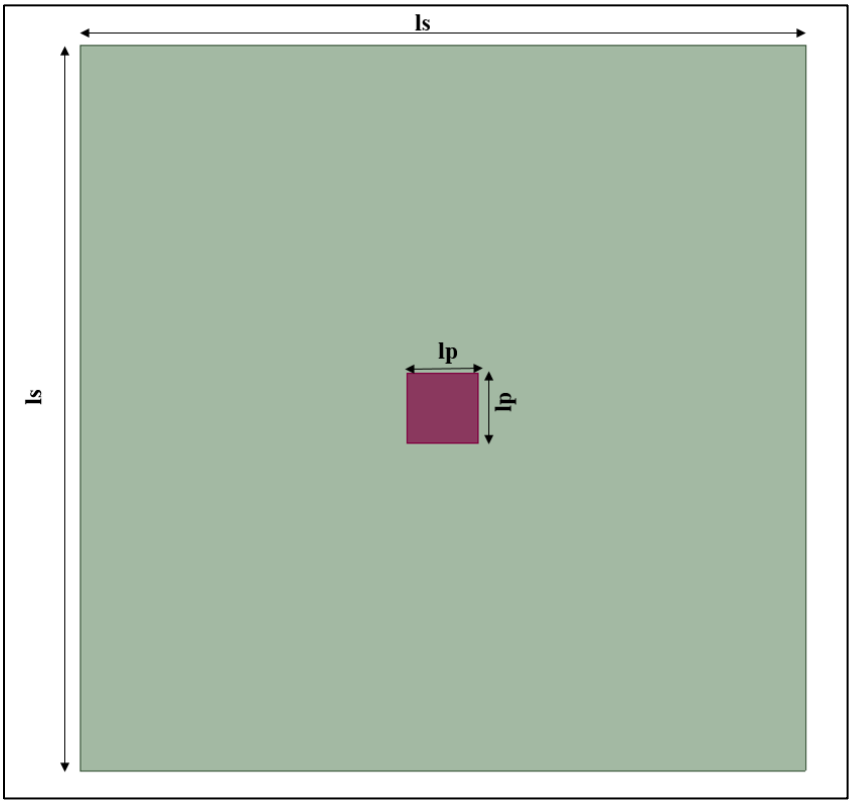


Fig.2. Radiating patch on the bottom face of middle substrate.

The substrate has dimensions of ls×ls. The radiating patch has dimensions lp×lp. The central placement of the patch and the substrate's characteristics are optimized for efficient radiation, low return loss, and specific resonant frequencies. The dimensions in the figure are specified in table 2.

Table 2: Dimensions of the patch

Parameters	**Description**	**Dimensions (mm)**
ls	Length of substrate	7.36
lp	Length of patch	0.716

Top substrate: For the third antenna topology (Fig.3), shows a Square-shaped loop of frequency selective surface superstrate layer that is designed and simulated on the bottom face of the top substrate with relative permittivity of 3 and thickness of 0.0639mm (Rogers RO3003). The FSS selectively allows or blocks certain frequencies of electromagnetic waves, based on its geometry and material properties.

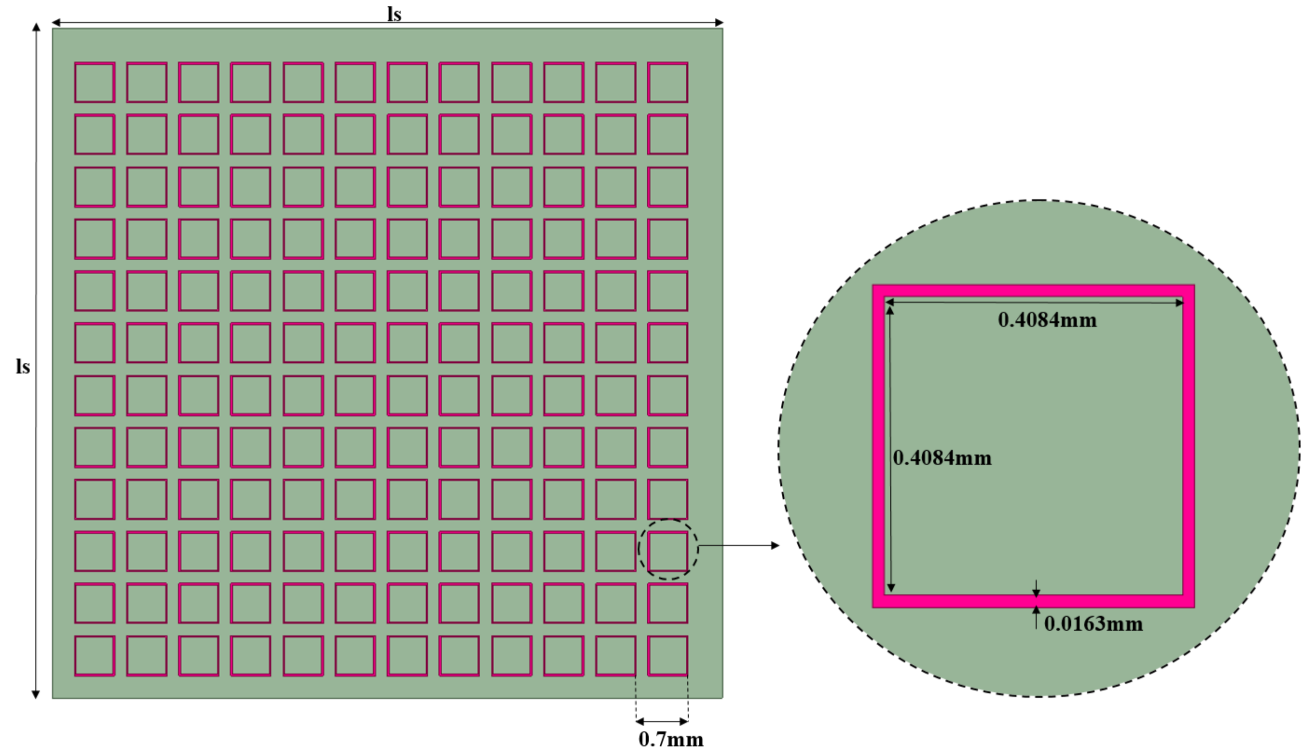


Fig.3. FSS superstrate layer on the bottom face of top substrate

The entire system put together looks such:

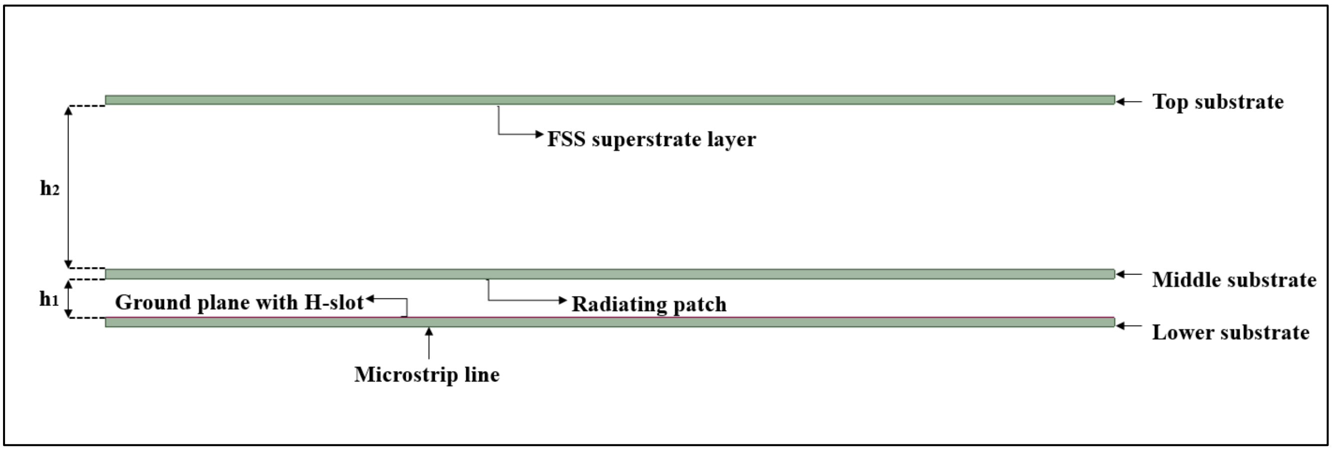


Fig. 4. Side-view of all the layers put together

The dimensions in the figure are specified in table 3.

Table 3: Dimensions of fig.4

Parameters	Description	Dimensions (mm)
h_1	Separation between top face of lower substrate and bottom face of middle substrate	0.286
h_2	Separation between top face of middle substrate and bottom face of top substrate	1.207

Results:

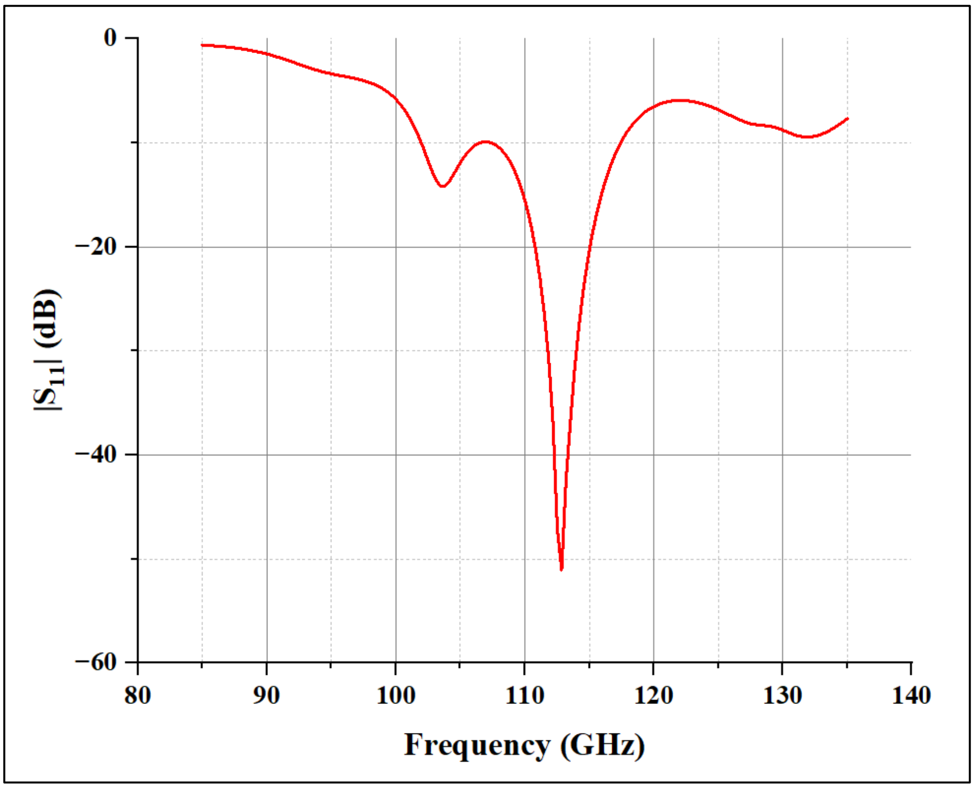


Fig. 5. Reflection coefficient vs frequency of the aperture coupled antenna with a FSS superstrate layer

The resonant frequency is obtained at 112.8 GHz with good coupling characteristics. Reflection coefficient of **-51 dB** is achieved.

The Gain vs frequency curve and the variation of VSWR with frequency is shown in fig. 6 and fig. 7 respectively.

From fig. 6, we can see that the **gain** of the antenna **at 112.8 GHz is 14.8 dB**.

From fig. 7, we observe that the **VSWR value at the resonant frequency is 1.0056** which gives us a near-perfect condition.

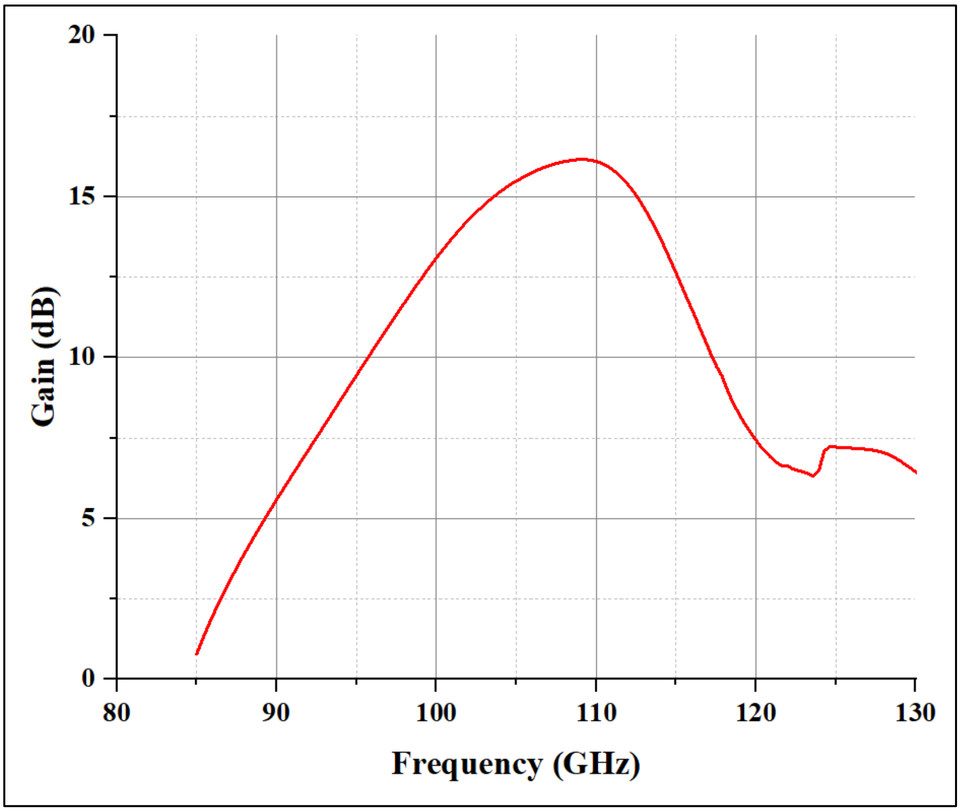


Fig. 6. Gain vs frequency of the aperture coupled antenna with a FSS superstrate layer.

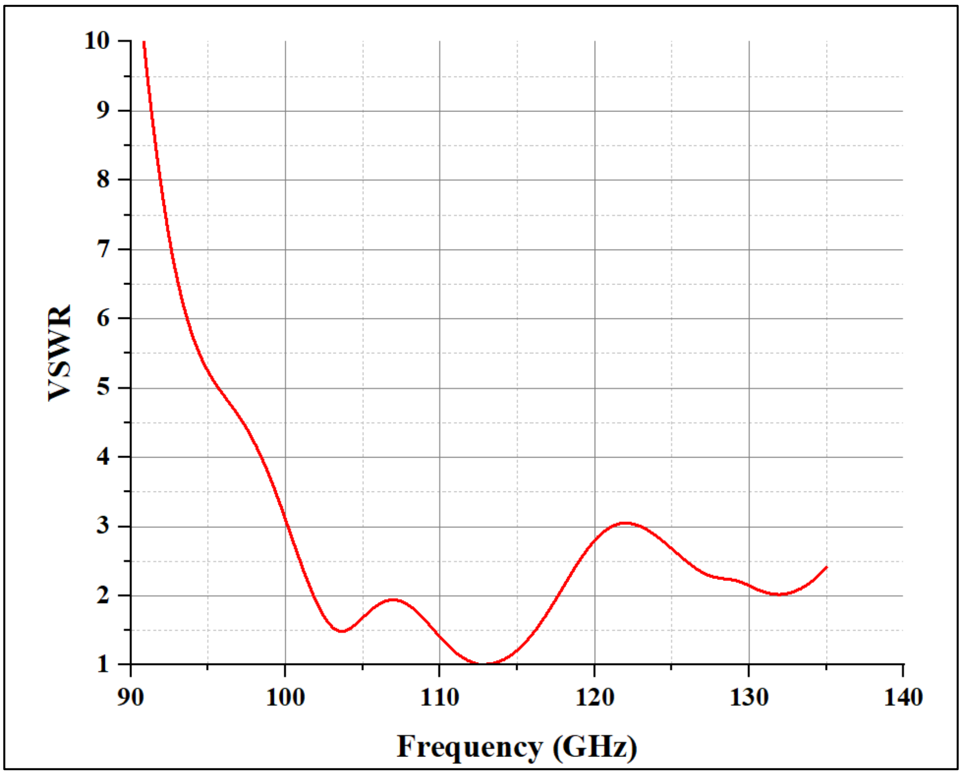
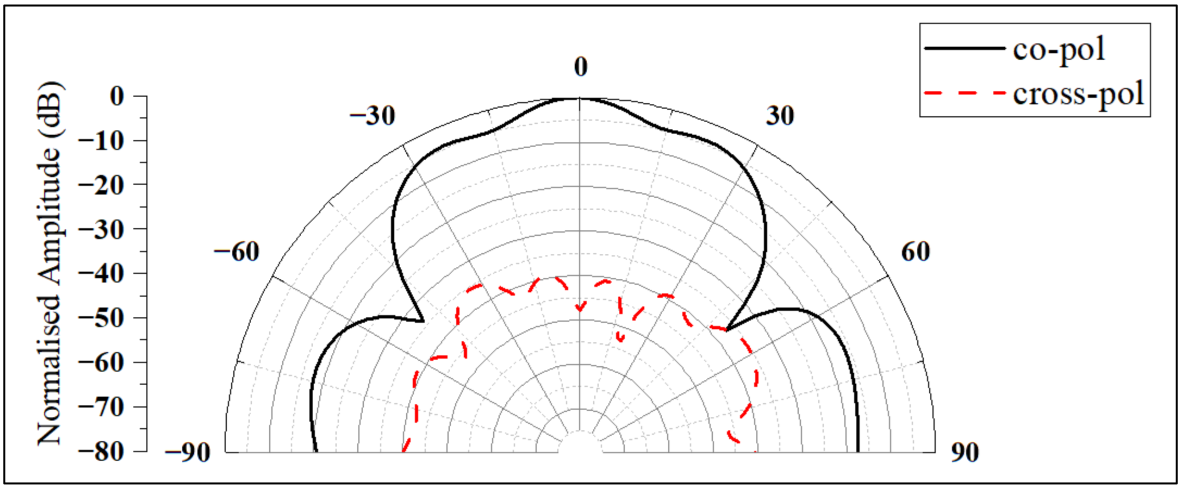
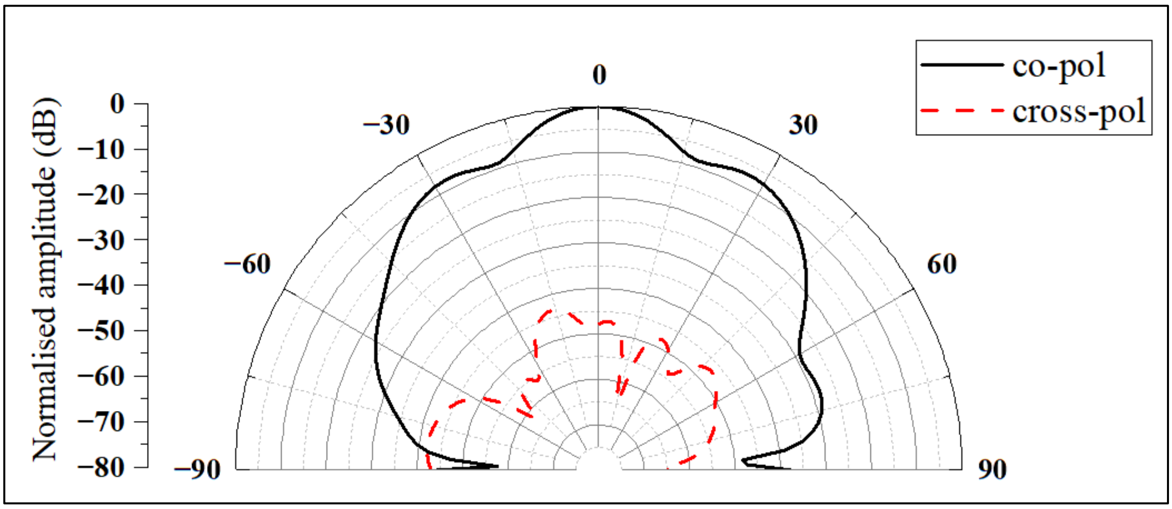


Fig. 7. VSWR vs frequency of the aperture coupled antenna with a FSS superstrate layer.

The radiation pattern of the Aperture-coupled microstrip antenna with FSS superstrate layer at the resonant frequency has been shown in fig. 8.



(a) ϕ = 0°



(b) ϕ = 90°

Fig. 8. Radiation pattern of the antenna at 112.8 GHz (a) ϕ = 0° (b) ϕ = 90°

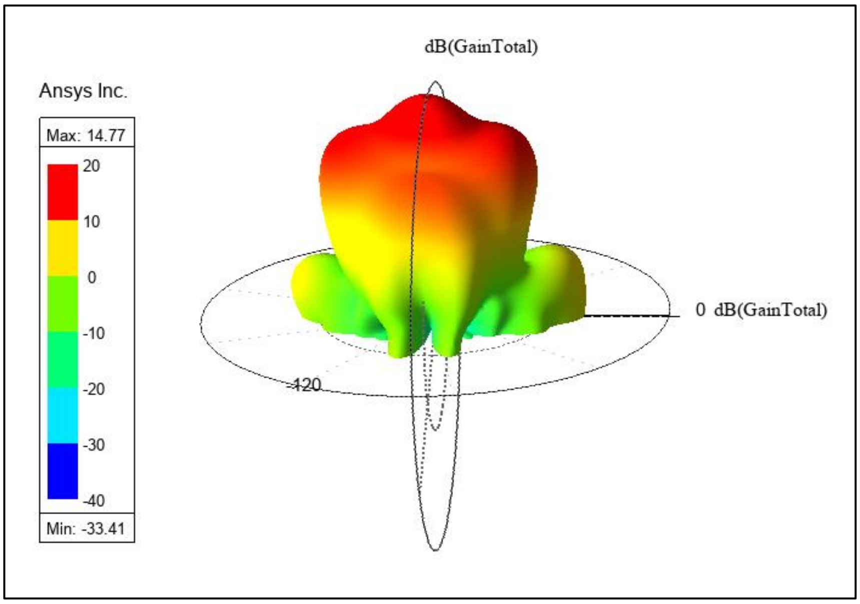


Fig. 9. 3-dimensional radiation pattern of the antenna at 112.8 GHz

The 3-dimensional radiation pattern of the aperture-coupled antenna with the FSS superstrate layer at the resonant frequency of 112.8 GHz gives us a **maximum gain of 14.77 dB**.